Shubham Gupta

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EDUCATION

MILA, Université Laval, Canada PhD in **Computer Science** (2023 - Sep. 2027, exp.)
Focus: reliable LLM systems, scalable retrieval/RAG, representation quantization, multimodal LLM adaptation, and privacy/safety evaluation for shared agentic systems

University of Waterloo, Canada M.Math. in **Combinatorics & Optimization** (2009-2011)
Thesis: *Building Networks in the Face of Uncertainty* GPA: 91.5%

Indian Institute of Technology, Delhi B. Tech. in **Computer Science** (2005-2009)
Thesis: *Approximation Algorithms for Facility Location Problems* GPA: 8.97

SELECTED PUBLICATIONS

HAIPS Workshop, COLM, 2026 *PiSAs: Benchmarking Contextual Integrity in Multi-User Agentic Systems*, ***S. Gupta****, N.M. Sepahvand, et al.*

ICML, 2026 *Hierarchical Retrieval at Scale: Bridging Transparency and Efficiency*, ***S. Gupta****, Z. Li, T. Chen, et al.*

ICML, 2025 *LAST SToP For Modeling Asynchronous Time Series*, ***S. Gupta****, T. Durand, G. Taylor, et al.*

Interspeech, 2026, Oral *Closing the Modality Gap via Simplex-Constrained Representations*, ***S. Gupta****, et al.*

Interspeech, 2025 *LiSTEN: Learning Soft Token Embeddings for Neural Audio LLMs*, ***S. Gupta****, et al.*

Interspeech, 2024 *Phoneme Discretized Saliency Maps for Explainable Detection of AI Generated Voice*, ***S. Gupta****, M. Ravanelli, P. Germain, et al.*

IPCO, 2010 *A 3-approximation for facility location with uniform capacities*, ***S.Gupta****, A. Aggarwal, L. Anand, et al. (Mathematical Programming, 2013 (an extended abstract appeared in IPCO, 2010))*

TECHNICAL SKILLS

LLM systems, privacy/safety evaluation, multi-user agentic systems, retrieval/RAG, dense retrieval, reranking, semantic search, hierarchical retrieval, representation quantization, semantic IDs, multimodal embeddings, PEFT, Python, PyTorch, Hugging Face Transformers.

ACADEMIC SERVICE

Assistant Chair, NeurIPS 2026 Evaluations & Datasets Track.

RESEARCH INTERNSHIPS

ServiceNow, Montreal (Mitacs with Université Laval) **November 2025 - June 2026**
Developed **PiSAs: Privacy in Shared Agentic Systems**, a benchmark of **85 manually curated enterprise-style scenarios** for evaluating privacy failures in multi-user LLM agent systems. PiSAs formalizes contextual-integrity evaluation with dual labels for **task appropriateness** and **per-user visibility**, measuring leakage across inter-agent communication, shared/private memory, and final outputs. Across **3 agent topologies**, **3 memory regimes**, and **3 LLM backbones**, multi-agent partitioning reduces violation rates by up to **50 points** for appropriateness and **70 points** for visibility, but all configurations still show $>$**75%** runs with at least one privacy violation.

ServiceNow, Montreal (Mitacs with Université Laval) **October 2024 - June 2025**
Developed **ReTreever** (**ICML 2026**), a hierarchical retrieval framework that is **160$\times$ faster than RAPTOR** with **+30 NDCG@10 on NQ**, while scaling to **124.5M documents** where ScaNN exceeded **1TB RAM**. ReTreever learns a binary tree over frozen encoder embeddings using **contrastive learning**, cross-attention split functions, and **probabilistic differentiable routing**, producing compact coarse-to-fine representations without LLM calls. The method improves over the base encoder, enables speed–accuracy tradeoffs at inference, and is validated across text, image, and speech retrieval benchmarks.

Amazon, California **June 2025 - October 2025**

Developed a generative retrieval approach for multimodal music search and recommendation at Amazon Music Search, achieving approximately **56% higher Recall@30** and **33% higher NDCG@30** than a standard semantic-ID quantization baseline in offline MovieLens-derived ablations. The method learns discrete **semantic IDs** via representation quantization over audio snippets, track metadata, and recommendation embeddings, unifying search and recommendation in a shared semantic code space for **efficient indexing and scalable retrieval**. The work is currently being prepared for submission.

Borealis A.I., Montreal **May 2024 - October 2024**

Developed **LASTS (Language-modeled Asynchronous Time Series)** (**ICML 2025**), an **LLM-based sequence modeling framework** that achieved **+13.55% average Macro-F1 over QLoRA** across forecasting, imputation, and anomaly detection tasks. LASTS reformulates **asynchronous time series modeling** as a **language modeling** problem, enabling one LLM-based architecture to handle multiple irregular-event sequence tasks. Introduced **Stochastic Soft Prompting (StoP)**, a parameter-efficient adaptation method that improves robustness through stochastic prompt truncation and outperforms vanilla soft prompting by **+12.69% average Macro-F1**.

PRIOR INDUSTRY EXPERIENCE

GOOGLE, Mountain View **Software Engineer** **2011 – 2014**

Knowledge Graph (KG) Improved fact-extraction pipelines from Wikipedia for Google's Knowledge Graph, reducing reconciliation losses and significantly expanding coverage. Designed methods for extracting unlinked entities, achieving **18% more valid extractions**, and implemented new heuristics that **increased recall tenfold** while maintaining comparable precision.

STRIPE, San Francisco **Senior Machine Learning Engineer** **2020 – 2021**

Merchant Fraud Team: Designed and enhanced fraud detection models for new merchant sign-ups, improving the existing anomaly detection framework to **reduce mean time to detect (MTTD) by 72%** and increase attacks caught by **45%**. Led a working group integrating text embeddings from **FastText** and **BERT**-based models with traditional NLP features such as **count** and **tf–idf vectorizers**, yielding a **5% improvement** in overall merchant fraud classification **ROC–AUC**.

DESCRIPT, San Francisco **Senior Applied Research Scientist** **2021 – 2023**

Performed research in **deep learning, self-supervised representation learning, and audio synthesis** to develop high-accuracy voice verification and zero-shot text-to-speech (TTS) systems. Designed and optimized a **zero-shot IPA-to-phoneme aligner** capable of multilingual training without speaker tuning, achieving a **mean word boundary alignment error of 23 ms** on an English golden set which was a **77% improvement** over baseline alignments, leading to substantial gains in TTS audio quality metrics. Additionally, proposed novel evaluation metrics to accelerate model iteration, guide architectural extensions, and enhance training efficiency.

SNAPCHAT, Los Angeles **Senior Software Engineer** **2014 – 2018**

Smart Content Precaching: Led a team that developed **machine-learning–driven personalized content preloading strategies**, resulting in a **100% improvement in utilization** of precached content. This work also achieved a **50% reduction** in loading screens, a **7% growth** in monthly active users (MAU), and a **5% increase** in overall content engagement.

RETENTION SCIENCE, Los Angeles **Senior Research Scientist** **2018 – 2020**

Led the Data Science team managing **30+ predictive ML models**, including recommendation systems, lead scoring, churn prediction, and CLV modeling. These models powered **4B+ daily predictions** for **385M users** across **145 e-commerce clients**, optimizing email marketing campaigns across various industries.